

# KRIS DCOSTA

San Diego, CA

+1 (858) 265-8381 — kdcosta@ucsd.edu — linkedin.com/in/kris-dcosta — github.com/KrisDcosta

## Education

University of California, San Diego | M.S. in Data Science

Sep 2025 – Jun 2027

GPA: 4.0/4.0

San Diego, CA

Sardar Patel Institute of Technology | B.Tech in ECE

Dec 2021 – Jun 2025

GPA: 3.95/4.0 Minor in Computer Engineering

Mumbai, India

Relevant Coursework: Machine Learning, Statistical Models, Efficient AI, Probability & Statistics, Linear Algebra

## Experience

Indian Institute of Technology Bombay

Jan 2024 – Jul 2025

Research Intern, Vision and Image Processing Lab

Mumbai, India

- **Led a 3-member team** to design and prototype an end-to-end computer-vision pipeline for photovoltaic module defect detection, reducing manual inspection time by 83% (from ~3 minutes to ~30 seconds) per module
- Built robust preprocessing and image-processing pipelines to handle noisy, real-world field data, including background clutter, illumination variation, and class imbalance
- Prepared, curated, and validated annotated datasets using CVAT, and explored model-assisted labeling strategies to improve annotation efficiency under limited labeling budgets
- Trained and evaluated lightweight CNN models in TensorFlow, benchmarking performance across architectures and tuning hyperparameters, analyzing failure cases to guide iterative model improvements

IBM SkillsBuild

Jun 2023 – Jul 2023

Data Analytics Intern (Project-Based)

Remote

- Built an end-to-end analytics workflow on financial time-series data, including EDA, feature engineering, XGBoost modeling, and dashboard-based reports to surface trends and anomaly insights for business interpretation

## Projects

ARMD-NetX: Multi-Modal Ensemble for ARMD Detection | Python, TensorFlow, OpenCV Aug 2024 – Jun 2025

- **Curated and unified a multi-modal retinal imaging dataset (~4,700 images)** by combining 8 public sources, addressing class imbalance and inconsistent formats across OCT and fundus imaging modalities
- Designed modality-specific preprocessing pipelines and trained CNN models, followed by ensemble modeling to improve robustness under limited data and compute constraints
- Achieved strong classification performance (OCT: F1  $\approx$  99%, Fundus: F1  $\approx$  98%) and applied Grad-CAM to interpret model attention and verify clinically relevant attention regions across imaging modalities

TagSense: Smartphone Based Gesture Detection | Python, NumPy, scikit-learn, Android(Java) Nov 2025 – Present

- **Designed and prototyped an end-to-end sensing and algorithm pipeline** for time-series gesture detection using multi-sensor smartphone data (accelerometer, gyroscope, gravity)
- Supported internal data collection by defining gesture protocols and collecting labeled sensor data from 30+ participants under real-world usage conditions
- Implemented sliding-window segmentation and engineered time and frequency-domain signal features to characterize motion patterns
- Integrated an initial rule-based detection system into an Android app, achieving strong usability in early testing (SUS=83) and analyzed false positives to inform subsequent ML-based iterations

Recipe Recommendation & Rating Prediction (Food.com) | Python, Pandas, Scikit-learn Nov 2025 – Dec 2025

- **Built and evaluated recommender algorithms on a large-scale, sparse real-world dataset (~1.1M user-item interactions)**, analyzing user-behavior patterns, robustness under data sparsity, and temporal drift
- Conducted EDA to analyze rating distributions, long-tail popularity, and temporal drift over 18 years of interactions
- Compared baseline, content-based, and latent factor models, extending matrix factorization with time-aware user bias terms to improve generalization (test RMSE  $\approx$  1.20)
- Evaluated the impact of ambiguous 0-star ratings and quantified trade-offs between metric optimization and real-world signal integrity

## Skills

**Programming & Data:** Python, SQL, R, Java, C++, Pandas, NumPy, scikit-learn

**Machine Learning:** PyTorch, TensorFlow, Supervised & Unsupervised Learning, Computer Vision, Multimodal Systems

**Statistics & Applied ML:** Statistical Modeling, Regression, Classification, Time-Series Analysis, Feature Engineering, Model Evaluation, Exploratory Data Analysis (EDA)

**Tools & Systems:** Git, Docker, Tableau, AWS & GCP, REST APIs

## Publications

**Selected Peer-Reviewed Publications:** IEEE ICCNT 2025; IEEE ICC 2025; IEEE INOACC 2025

**Patent:** DoppelScan — Dual Authentication Contactless Attendance System (Indian Patent Office, 2024)